15.465 Final Project Report

Data File: fmreg_data_std_98.csv

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Executive Summary

Building on the Part 1 multivariate findings, **we recommend a Buy on Johnson & Johnson, Neutral on Pfizer, and Sell on Novo Nordisk.** All ten Part 1 signals were statistically significant at the 10 percent level, so all ten enter our composite. We rank the three firms on each signal in the direction implied by its Part 1 multivariate coefficient and combine the ranks two ways: an unweighted sum and a t-statistic-weighted sum. Both methods produce the same Buy / Neutral / Sell ordering, so the call is robust to the choice of weighting. Johnson & Johnson scores best on the four signals with the largest absolute t-statistics (Insider Selling, Volatility, Momentum, Earnings Surprise), Novo Nordisk scores worst on three of the same four, and Pfizer sits in the middle. The recommendation is consistent with Part 1's Sharpe ratio analysis, which identified those same four signals as the strongest predictors of monthly returns.

Data Analysis

Univariate Analysis

Of the 10 signals analyzed via iterative monthly cross-sectional regressions of returns, momentum, earnings surprise, and SUE have the strongest positive relationships with returns over time. These factors also are the factors with the highest positive percentage of beta coefficients over the 60-month period. Insider selling and volatility, on the other hand, have the strongest average negative relationships with returns over time, also possessing the lowest positive fraction of beta coefficients over the same period. Overall, insider selling has the strongest relationship with returns and the average R^2 for its coefficient is more than double the next highest R^2 of any other factor's beta, indicating the explanatory power of insider selling to predict returns.

Table 1 Average Monthly Univariate Coefficients.

Factor	Average Beta	T-Statistic	Average R^2	Average Number of Observations
Momentum	0.0157	7.0353	0.0447	200
Earnings Surprise	0.0155	9.8548	0.0195	200
SUE	0.0138	7.6186	0.0210	200
Accruals	0.0077	3.8267	0.0116	200
Gross Profitability	0.0042	2.7054	0.0071	200
Book-To-Market	0.0033	2.2794	0.0055	200
F-Score	0.0029	1.5718	0.0089	200
External Financing	-0.0084	-4.5547	0.0121	200
Volatility	-0.0143	-6.2233	0.0240	200
Insider Selling	-0.0274	-7.3429	0.1000	200

Table 2 Positive Fraction of Monthly Univariate Coefficients.

Factor	**Positive Fraction of Coefficients**
Momentum	0.9000
Earnings Surprise	0.8333
SUE	0.8000
Accruals	0.7167
F-Score	0.6833
Gross Profitability	0.6667
Book-To-Market	0.6500
External Financing	0.2667
Volatility	0.2667
Insider Selling	0.2167

There is variability in the estimates for all coefficients over time, but the mean for all factors but insider selling, momentum, and volatility seems to remain roughly constant. For f-score, gross profitability, and accruals, the volatility of the coefficient seems to increase around 2025, though the average beta value remains stable throughout. The momentum coefficient is higher towards the beginning of the 60-month period on average than at the end of it, though a large spike is present around 2025. Moreover, the insider selling beta seems to follow the inverse pattern of the momentum coefficient and on average increases throughout the 60-month period. The volatility beta itself is quite volatile, though it generally dips a bit in 2023 from its earlier levels and then increases past its 2021 level in 2025 and 2026.

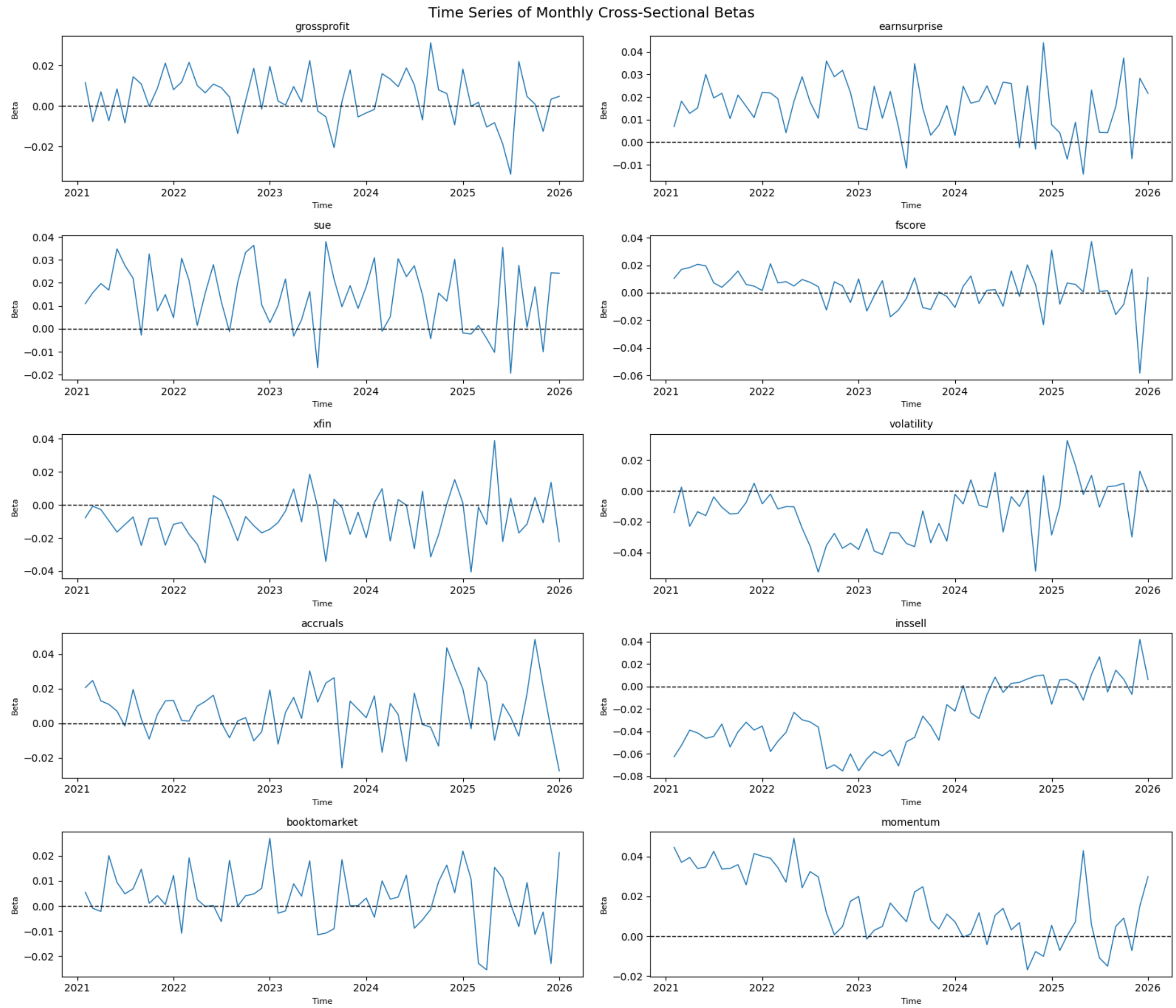

Figure 1 Time Series of Monthly Univariate Coefficients.

Other than f-score, all factors' coefficients are statistically significant[1]. The insider selling signal likely works because members of the c-suite typically sell their shares when they possess information indicating the price may fall. The momentum signal is likely significant because stocks that are doing well often continue to do well in a short-term horizon, and because investors generally underreact to good news of a stock's performance over this period. The earnings surprise and SUE signals, which are quite related, likely work for a similar reason – because the market often underreacts to good news so prices will not immediately reflect a positive earnings announcement. The volatility signal is likely significant due to investors' preference for lottery-like stocks leading them to overpay for more volatile stocks. The accruals and gross profitability signals likely work because of investor tendencies to focus on simple bottom-line metrics like EPS in an earnings report, causing them to neglect to consider signals that may signify either future growth (accruals) or efficiency at generating value from assets (gross profitability). The book-to-market signal likely is significant because investors tend to overextrapolate positive performance

[1] Signals were judged as statistically significant if their t-statistic exceeded 1.645 in absolute value.

of low B/M (high price) firms, leading them to neglect high (B/M) firms. Finally, the external financing signal is likely significant because managers tend to raise capital when they believe the stock price is too high and will thus fall in the future.

Multivariate Analysis

All signals that were significant predictors of return in the univariate regressions remain significant predictors of return in the multivariate analysis. Moreover, f-score, the only insignificant factor in the univariate analysis, becomes significant in the multivariate tests.

Table 3 Average Monthly Multivariate Coefficients.

Factor	**Average Beta**	**T-Statistic**	**Average R^2**	**Average Number of Observations**
Momentum	0.0147	6.1364	—	—
Earnings Surprise	0.0114	5.9413	—	—
Accruals	0.0083	4.5543	—	—
SUE	0.0052	2.5663	—	—
Gross Profitability	0.0042	2.7846	—	—
F-Score	0.0038	2.1132	—	—
Book-To-Market	0.0032	2.4243	—	—
External Financing	-0.0096	-5.5561	—	—
Volatility	-0.014	-6.184	—	—
Insider Selling	-0.0268	-7.3305	—	—
Model	—	—	0.2252	200

The t-statistics and coefficients are nearly identical for volatility, insider selling, gross profitability, and B/M across the two analyses, indicating that these factors are robust independent indicators of returns. The t-statistics and betas for earnings surprise and SUE decrease notably in the multivariate case compared to the univariate tests, despite earnings surprise still being the second-largest positive coefficient in the multivariate setting. As SUE and earnings surprise are highly correlated – both measure how much recent earnings deviate from expected earnings – this decrease in coefficients and t-stats is unsurprising when both factors are controlled for within the same regression. The momentum t-statistic and coefficient also decrease in the multivariate case, but more moderately than SUE and earnings surprise. The t-statistics and betas for both f-score and accruals increased in the multivariate tests, indicating the predictive power of these signals may only be uncovered when correlated signals are controlled for. The monthly coefficient estimates for the multivariate regression generally follow similar trends to the coefficients in the univariate regressions.

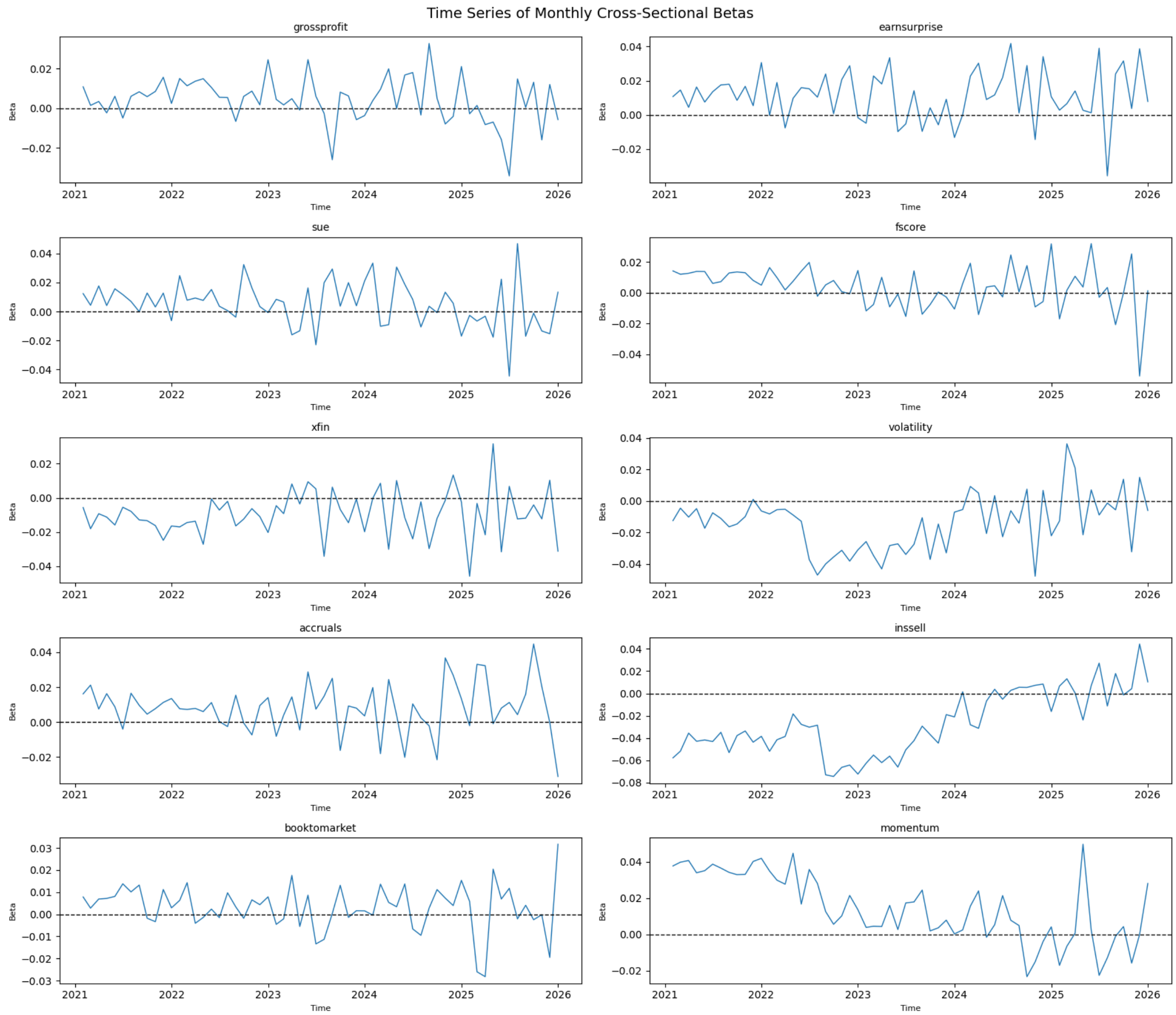

Figure 2 Time Series of Monthly Multivariate Coefficients.

The average R^2 of the model is 0.225. This value is more than double the highest R^2 from a single factor in the univariate tests, 0.1 for insider selling. Moreover, this value is more than an order of magnitude higher than many of the R^2 values for other univariate factors. This comparison shows that there is significantly more predictive power when the signals are combined in the multivariate model and that these tests can explain a non-trivial share of cross-sectional return variation.

All coefficients drawn from the multivariate regression represent the predicted monthly return premium for a firm one standard deviation above the cross-sectional mean on a given signal, holding all other signals constant. Positive premiums include momentum (+1.5%), earnings surprise (+1.1%), accruals (+0.8%), SUE (+0.5%), gross profitability (+0.4%), f-score (+0.4%), and book-to-market (+0.3%). Negative premiums include external financing (−0.1%), volatility (−1.4%), and insider selling (−2.6%).

Sharpe Ratio

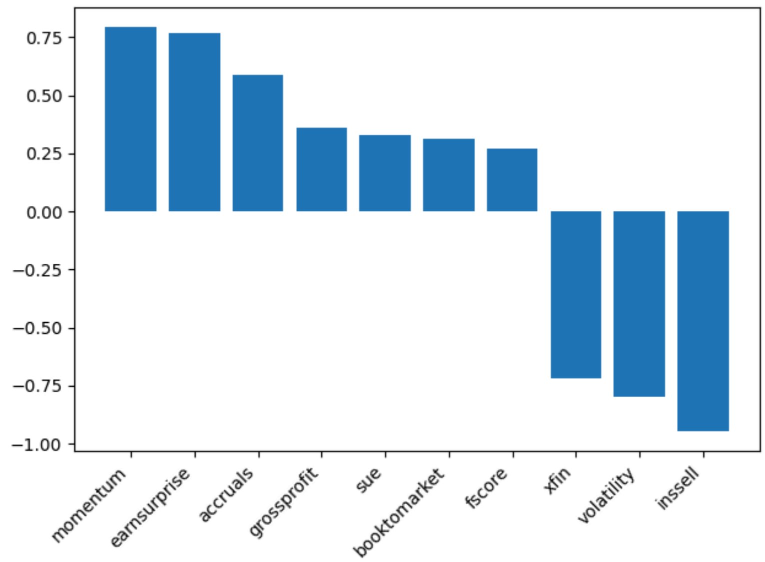

Figure 3 Sharpe Ratios of Monthly Multivariate Coefficients.

Of the 10 factors in the multivariate analysis, insider selling, volatility, momentum, and earnings surprise offer the strongest performance in terms of Sharpe ratio. As to be expected, momentum and earnings surprise have positive Sharpe ratios while insider selling and volatility have negative Sharpe ratios.

The implication from the absolute value of these Sharpe ratios is that the four factors listed above consistently predict returns each month more so than any of the other signals. This motivates the choice of an investment strategy of buying stocks with higher-than-average signals of momentum and earnings surprise, based on the positive Sharpe ratios, and shorting stocks with higher-than-average signals of insider selling and volatility, based on the negative Sharpe ratios. Since the Sharpe ratio reflects consistency of returns prediction, it naturally informs how these signals should be weighted when combined into a composite strategy, with each signal's independence ensuring that combination reduces reliance on any single predictor.

Research & Recommendation

Signal Collection and Measurement

For each of the three pre-registered companies, we collected the most recent value of every signal that was statistically significant in the Part 1 multivariate regression. All ten signals cleared the 10 percent significance threshold, so all ten are reported. Financial inputs were sourced from each company's most recent SEC filing (10-K or 20-F for annual data, 8-K or 6-K for quarterly earnings, Form 4 or Article 19 PDMR notifications for insider transactions), and price and return inputs were sourced from Yahoo! Finance, with WRDS IBES used for analyst consensus where applicable. The end-of-April 2026 cutoff date specified in the assignment was applied throughout. The three sub-sections below present each company's signal table and discuss any deviations from the Part 1 definitions or any treatment of missing or ambiguous data.

Pfizer (PFE)

Table 4 Statistically Significant Signals for PFE.

Signal	**Value**	**Source**	**As-Of**
Insider Selling	0	SEC Form 4 filings	4/27/26
Book-To-Market	0.6129	10K for the fiscal year ending 12/31/25; Yahoo! Finance	12/31/25
Momentum	-0.1237	Yahoo! Finance	3/31/26
Gross Profitability	0.2234	10K for the fiscal year ending 12/31/25	12/31/25
Earnings Surprise	-0.0112	8K earnings release filed 2/3/26; Yahoo! Finance	2/3/26
SUE	-0.0085	8K earnings release filed 2/3/26; Yahoo! Finance	2/3/26
F-Score	5	10K for the fiscal year ending 12/31/25	12/31/25
External Financing	-0.0004	10K for the fiscal year ending 12/31/25	12/31/25
Volatility	0.0159	Yahoo! Finance	4/30/26
Accruals	-0.0189	10K for the fiscal year ending 12/31/25	12/31/25

The calculation of all signals for PFE were carried out according to the formulas in Part 1. However, two signals warrant additional explanation. For insider selling, all SEC Form 4 filings submitted by Pfizer Directors and relevant Officers (CEO, CFO, COO) during the trailing three-month window ending April 27, 2026 were reviewed. No open market purchase (code "P") or sale (code "S") transactions were identified in Table I of any filing. All entries consisted of routine compensation grants (code "A") and tax withholding on vesting (code "F"). Accordingly, the signal is set to zero.

For external financing, no explicit proceeds from common stock issuance line items appeared in the financing activities section of the 10-K as of 12/31/2025. Per Note 12, Pfizer did not repurchase any shares during any period presented. Net equity issuance was therefore set to zero, and external financing reflects net debt issuance only.

Novo Nordisk (NVO)

Table 5 Statistically Significant Signals for NVO.

Signal	**Value**	**Source**	**As-Of**
Insider Selling	0.0000060	Article 19 PDMR notifications	2/10/26
Book-To-Market	0.1602	20F for the fiscal year ending 12/31/25; Yahoo! Finance	4/30/26
Momentum	-0.6085	Yahoo! Finance	3/31/26
Gross Profitability	0.4610	20F for the fiscal year ending 12/31/25	12/31/25
Earnings Surprise	-0.0008	6K earnings release filed 2/3/26; WRDS IBES; Yahoo! Finance	2/3/26
SUE	-0.0008	6K earnings release filed 2/3/26; WRDS IBES; Yahoo! Finance	2/3/26
F-Score	5	20F for the fiscal year ending 12/31/25	12/31/25
External Financing	0.0430	20F for the fiscal year ending 12/31/25	12/31/25
Volatility	0.0333	Yahoo! Finance	4/30/26
Accruals	-0.0307	20F for the fiscal year ending 12/31/25	12/31/25

The calculation of all signals for NVO followed the formulas in Part 1. Several deviations warrant additional explanation given Novo Nordisk's status as a Danish foreign private issuer (FPI). The equivalent disclosures to Pfizer's 10K and 8K are the Form 20F (filed with the SEC on 2/4/26 for fiscal year 12/31/25) and the Form 6K covering the Q4 2025 earnings release dated 2/3/26. All financials are in DKK under IFRS, and the line items map directly to the Part 1 definitions.

For insider selling, Section 16 does not apply to FPIs, so no Form 4 filings exist. Insider transactions are disclosed under Article 19 of EU MAR (Regulation 596/2014), with notifications published as company announcements via Globenewswire and the Danish FSA's OAM database. All Novo Nordisk Article 19 PDMR notifications between 1/30/26 and 4/30/26 were reviewed. Consistent with the Form 4 treatment applied for Pfizer, only Article 19 entries that represent open market purchases or sales were counted. Share remuneration grants and tax withholding on vesting were excluded. Only one notification in the trailing three month window involves a CEO, CFO, COO, or Director, an open market sale of 26,557 B shares by CFO Karsten Munk Knudsen on 2/10/26 across two execution venues. The signal value is this single sale divided by total shares outstanding of 4.448 billion at 12/31/25 (per the 20F share capital disclosure of 4.51 billion shares issued less treasury holdings).

For Earnings Surprise and SUE, the calculation for an IFRS DKK reporter uses DKK on both sides of the ratio. Quarterly EPS in DKK is the numerator and the NOVO B Copenhagen close in DKK is the denominator. WRDS IBES had not yet ingested the Q4 2025 actual at the time of measurement, so Q4 2025 EPS was recovered from the FY 2025 reported diluted EPS of 23.03 DKK in the 2/3/26 release less the IBES Q1 to Q3 actuals of 16.99 DKK, giving 6.04 DKK. The Q4 2025 consensus comes from the latest IBES annual mean before the announcement (statpers 1/15/26, 23.33 DKK across 19 estimates) less the same Q1 to Q3 actuals, giving 6.34 DKK. The Q4 2024 actual of 6.34 DKK is from IBES directly. Both Earnings Surprise and SUE land at -0.0008 because the imputed Q4 2025 consensus and the Q4 2024 actual are both 6.34 DKK. Analyst expectations for the quarter had been anchored on the prior year print.

For Book-To-Market, both numerator and denominator are in DKK. Book equity is the 12/31/25 total equity of 194,047 DKK million from the 20F consolidated balance sheet, which equals total assets of 542,902 DKK million less total liabilities of 348,855 DKK million. Market capitalization is the NOVO B Copenhagen close on 4/30/26 of 272.25 DKK multiplied by total shares outstanding of 4.448 billion.

For External Financing, unlike Pfizer's near zero result, Novo Nordisk's 2025 financing activity is dominated by a 103,931 DKK million long term debt issuance per the consolidated cash flow statement. Repurchase of own shares of 1,388 DKK million and long term debt repayment of 79,188 DKK million net to a positive External Financing value, opposite in sign to Pfizer's -0.0004. No new equity was issued in either fiscal year.

The F-Score follows the Piotroski (2000) specification, with the leverage criterion measured as the change in the long term debt to total assets ratio. Three of the nine criteria fail in 2025. ROA declined relative to 2024, the leverage ratio rose with the increase in long term debt, and gross margin contracted as cost of goods sold grew faster than revenue.

Johnson & Johnson (JNJ)

Table 6 Statistically Significant Signals for JNJ.

Signal	Value	Source	As-Of
Insider Selling	0	SEC Form 4 filings	4/30/26
Book-To-Market	0.1474	10K for the fiscal year ending 12/28/25; Yahoo! Finance	4/30/26
Momentum	0.3517	Yahoo! Finance	3/31/26
Gross Profitability	0.3210	10K for the fiscal year ending 12/28/25	12/28/25

Earnings Surprise	0.0001	8K earnings release filed 1/21/26; WRDS IBES; Yahoo! Finance	1/21/26
SUE	0.0020	8K earnings release filed 1/21/26; WRDS IBES; Yahoo! Finance	1/21/26
F-Score	3	10K for the fiscal year ending 12/28/25	12/28/25
External Financing	0.0243	10K for the fiscal year ending 12/28/25	12/28/25
Volatility	0.0106	Yahoo! Finance	4/30/26
Accruals	0.0114	10K for the fiscal year ending 12/28/25	12/28/25

The calculation of all signals for JNJ followed the formulas in Part 1. Johnson & Johnson is a US domestic issuer listed on the NYSE, so the disclosure regime and reporting standard match the Pfizer methodology. Three signals warrant additional explanation.

For insider selling, all SEC Form 4 filings for CIK 0000200406 between 1/30/26 and 4/30/26 were reviewed. Following the 2018 reorganization, Johnson & Johnson has no single Chief Operating Officer, so the search was restricted to CEO Joaquin Duato, CFO Joseph Wolk, and the Board of Directors. Two Director filings appear in the window, both routine compensation grants under the Deferred Fee Plan for Directors. Eugene A. Woods received 154.869 DSUs on 3/12/26 and Daniel E. Pinto received 1,712 cash settled DSUs on 4/23/26. Consistent with the Pfizer treatment, code A grants and code F tax withholding entries were excluded. Only open market code P or code S transactions count, of which none appear in the window. The signal is therefore zero.

For Earnings Surprise and SUE, the IBES adjusted EPS series is used on both the actual and consensus sides, consistent with the street EPS basis tracked by analysts. The Q4 2025 actual of $2.46 is from IBES directly. The implied Q4 2025 consensus of $2.43 is the latest IBES annual mean before the announcement (statpers 1/15/26, $10.77 across 13 estimates) less the IBES Q1 to Q3 2025 actuals ($2.77 + $2.77 + $2.80 = $8.34). The Q4 2024 actual of $2.04 is from IBES directly.

For Accruals, the convention used across the three companies computes (NI − CFO) / total assets on a fiscal year annual basis rather than the literal "most recent fiscal quarter" wording in the Part 1 specification. JNJ's positive Accruals value is mechanically driven by FY 2024 net income being depressed by approximately $7 billion in talc litigation charges. On a normalized basis the value would be closer to zero. External Financing similarly includes only long term debt activity, again following the team's convention. JNJ's short term debt issuance of $14.6 billion and repayment of $12.3 billion net to a small positive amount that would not change the sign of the signal.

Composite Scoring Methodology

We require a systematic combination of the ten Part 1 signals into a single composite score per firm, with the highest-scoring firm designated Buy, the middle Neutral, and the lowest Sell. Three inputs must be incorporated: the sign and magnitude of each Part 1 multivariate coefficient, the statistical reliability of each coefficient (its t-statistic), and the signal values calculated for Pfizer, Novo Nordisk, and Johnson & Johnson in Step A. We employ two methods that each capture all three inputs and run them in parallel as a robustness check on the recommendation.

Method 1: Direction-aware rank sum

For each signal, the three firms are ranked from 1 (worst for return) to 3 (best for return). "Best for return" means the highest signal value if the Part 1 coefficient is positive, and the lowest signal value if the coefficient is negative. The composite score is the unweighted sum of the ten ranks. The firm with the highest sum is Buy, the lowest is Sell, and the middle is Neutral. This method captures sign through the direction of the ranking, magnitude through the rank order itself (since rank already orders firms by their relative signal strength), and signal reliability through the binary inclusion criterion: every signal in Part 1 cleared the 10 percent significance threshold and is therefore counted with equal weight.

Method 2: t-statistic-weighted rank sum

Same direction-aware ranking as Method 1, but each rank is weighted by the absolute value of the Part 1 multivariate t-statistic before summing. This puts proportionally more weight on the four signals the Part 1 Sharpe analysis flagged as the strongest predictors (Insider Selling at |t|=7.33, Volatility at 6.18, Momentum at 6.14, and Earnings Surprise at 5.94) and proportionally less on F-Score, Book-To-Market, SUE, and Gross Profitability, which are significant but less powerful. This method captures the same three inputs as Method 1 but uses t-stat weights rather than equal weights to translate the strength of evidence from Part 1 into the composite.

Running both methods in parallel is a transparent way to demonstrate that the recommendation does not depend on an arbitrary choice of weighting. Method 1 is the simple ranking approach the assignment hints at. Method 2 is the more sophisticated version that uses Part 1's t-statistics as an explicit weighting scheme.

Composite Score Table

Direction-aware ranks across the three firms, computed from the Step A signal tables and the Part 1 multivariate coefficients. Rank 3 is best for predicted return.

Table 7 Composite Scores for PFE, NVO, and JNJ.

Signal	**β sign**	**PFE**	**NVO**	**JNJ**	**|t|**
Insider Selling	−	2.5	1	2.5	7.33
Volatility	−	2	1	3	6.18
Momentum	+	2	1	3	6.14
Earnings Surprise	+	1	2	3	5.94
External Financing	−	3	1	2	5.56
Accruals	+	2	1	3	4.55

Gross Profitability	+	1	3	2	2.78
SUE	+	1	2	3	2.57
Book-To-Market	+	3	2	1	2.42
F-Score	+	2.5	2.5	1	2.11
Method 1 sum		**20.0**	**16.5**	**23.5**	
Method 2 t-weighted		**92.6**	**65.2**	**115.7**	

Both methods agree. **The recommendation is Buy Johnson & Johnson, Neutral Pfizer, Sell Novo Nordisk.**

Justification

The drivers behind the result are concentrated in the highest-|t| signals identified in Part 1. Johnson & Johnson has the strongest Momentum (+0.3517), the lowest Volatility (0.0106), and the only positive Earnings Surprise (0.0001) and SUE (0.0020) values among the three firms. With Part 1 multivariate coefficients of +0.0147 on Momentum, −0.014 on Volatility, +0.0114 on Earnings Surprise, and +0.0052 on SUE, JNJ scores well on four of the five signals with the largest absolute t-statistics in the multivariate regression. Novo Nordisk sits at the opposite end on three of these four high-|t| signals: it has the worst Momentum (−0.6085), the highest Volatility (0.0333), and a positive External Financing reading of 0.0430 driven by the Akero Therapeutics acquisition, which pushes expected returns lower because External Financing has a Part 1 coefficient of −0.0096. NVO's Earnings Surprise and SUE are both modestly negative at −0.0008. Pfizer sits between the two on the highest-|t| signals: its Insider Selling is tied at zero with JNJ, its Volatility of 0.0159 sits between JNJ's 0.0106 and NVO's 0.0333, and its Momentum of −0.1237 sits between JNJ's +0.3517 and NVO's −0.6085. Pfizer's strongest contribution to its composite score is its Book-To-Market of 0.6129, by far the highest of the three firms, which reflects its depressed valuation. However, with a Part 1 t-statistic of only 2.42, Book-To-Market carries modest weight in Method 2 and cannot offset Pfizer's negative Earnings Surprise, negative SUE, and negative Accruals readings on the higher-|t| signals.

Sensitivity and Caveats

Three sensitivities warrant flagging. First, both composite methods agree on the Buy / Neutral / Sell ordering, so the call is not driven by the choice of weighting. Second, the Johnson & Johnson F-Score of 3 is mechanically inflated by FY 2024 net income being depressed by approximately 7 billion dollars in talc litigation charges; on a normalized basis the F-Score would be closer to Pfizer's 5 and Novo Nordisk's 5, which would not change the ordering since F-Score carries the lowest weight (|t|=2.11) of the ten signals. Third, the Pfizer and Novo Nordisk Insider Selling values are essentially zero (Pfizer exactly zero, Novo Nordisk 0.0000060) and tied at rank 2.5; ranking them as a tie is the most defensible treatment because the difference is far below any reasonable measurement precision.

Discretionary Analysis

The signals miss several real-world factors that matter here.

In April 2026, the White House imposed 100% tariffs on patented drugs but exempted firms that agreed to lower US prices and build US factories [1]. Pfizer, Novo Nordisk, and Johnson & Johnson all signed deals. The price cuts and factory spending will hit margins starting 2026, after our signal window.

For Pfizer, the high Book-To-Market of 0.6129 looks like value but reflects a depressed price tied to the post-COVID revenue cliff and the slow payoff from the $43 billion Seagen and $10 billion Metsera deals. A transition discount, not a bargain.

For Novo Nordisk, two signals were misleading. The negative External Financing reading comes from debt that funded the Akero Therapeutics deal, which expands the obesity and MASH pipeline. The Earnings Surprise of −0.0008 is partly mechanical: IBES had not yet recorded the Q4 2025 actual, so the consensus was reconstructed from older estimates and reflects analyst inertia. More importantly, in February 2026, CagriSema lost a head-to-head trial against Lilly's Zepbound, and Lilly now controls roughly 60% of the US GLP-1 market [2] [3].

For Johnson & Johnson, the F-Score of 3 is dragged down by roughly $7 billion in FY 2024 talc litigation charges; normalized, it sits closer to peers at 5. A Texas court rejected J&J's talc settlement plan in April 2025 for the third time, with possible settlement costs of $10–15 billion. Stelara sales fell 40% in the first nine months of 2025 [4].

Quant funds trust the model to average out such surprises. Discretionary funds would adjust: Novo's trial loss as a real setback, Pfizer as a value trap, and JNJ's lawsuit risk as already priced in.

Sources:

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